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Exploring User Adoption of ChatGPT: A Technology Acceptance Model Perspective

Jiaojiao Ma*, Pengcheng Wang*, Benqian Li, Tian Wang, Xiang Shan Pang, and Dake Wang

School of Media and Communication, Shanghai Jiao Tong University, Shanghai, China

ABSTRACT

In the rapidly evolving landscape of technology, the emergence of Chat Generative Pre-trained Transformer (ChatGPT) marks a pivotal milestone in the realm of Artificial Intelligence (AI). However, little research has reported the predictors of people's intentions to use ChatGPT. This pioneering study empirically examines user adoption through the lens of the Technology Acceptance Model (TAM) using a convenience sampling method. The study surveyed 784 ChatGPT users in China, of whom 58.93% were males. The results have revealed several key findings: (1) perceived usefulness, perceived ease of use, behavioral intention, and use behavior were positively correlated with each other; (2) behavioral intention acted as a mediating factor in the relationship between perceived usefulness and use behavior, as well as the relationship between perceived ease of use and use behavior; (3) perceived usefulness and behavioral intention played a chain-mediated role between perceived ease of use and use behavior; (4) the relationship between behavioral intention and use behavior exhibited greater strength among females compared to males; (5) the association between behavioral intention and use behavior was found to be stronger among urban users in comparison to their rural counterparts; (6) the connections between perceived ease of use and perceived usefulness, perceived ease of use and behavioral intention, and behavioral intention and use behavior were observed to be stronger among individuals with higher educational backgrounds relative to those with lower educational backgrounds. These findings provide crucial nuanced insights to advance the practical application of ChatGPT, emphasizing the need for enhanced usability and ease of use. However, this study exclusively captured usage behaviors within the Chinese user base. Future investigations could encompass diverse demographics across multiple countries, enabling cross-cultural comparisons.

KEYWORDS

Technology acceptance model (TAM); ChatGPT; perceived usefulness; perceived ease of use; behavioral intention; use behavior

1. Introduction

Artificial intelligence (AI) has significantly impacted various aspects of human life, seamlessly integrating into daily tasks and decision-making processes. ChatGPT, introduced by OpenAI in November 2022, marks a significant stride in Artificial intelligence's evolution. Its robust conversational abilities enable prompt and comprehensive responses to diverse user queries, leading to a swift accumulation of over 100 million users within two months. Building upon ChatGPT's remarkable capabilities, ongoing research endeavors are exploring its vast potential in various domains. Firstly, studies are exploring ChatGPT's potential applications in various fields like healthcare, tourism, chemistry, and psychology (Cascella et al., 2023; Uludag, 2023). Secondly, research is exploring the positive impact of ChatGPT adoption on organizations, societies, and individuals in terms of increased productivity while also addressing concerns related to privacy and security (Malik et al., 2023; Yao Tian et al., 2023). Thirdly, there is a call to identify key research questions regarding ChatGPT as a disruptive technology, including discussions on the benefits of embracing AI and establishing accountability rules (van Dis et al., 2023; Maad & Aljanabi, 2023), as well as addressing potential academic ethical issues (M. Mijwil et al., 2023). In terms of research methodology, there is a lack of quantitative research that objectively and systematically measures users' usage behaviors. Considering the explosive popularity of ChatGPT and its advanced capabilities compared to other publicly available chatbots, this study aims to bridge the existing research gap by focusing on measuring users' willingness to accept and the actual adoption of ChatGPT.

Recent research extensively examines the factors influencing AI adoption, primarily utilizing survey, experimental, and qualitative interview methods. These studies reveal key factors affecting user adoption. Firstly, AI technology characteristics, including anthropomorphism, intelligence, and reactivity, significantly influence adoption (Fernandes & Oliveira, 2021; Lu et al., 2019; Wu et al., 2023). Secondly,

CONTACT Dake Wang dakewang@sjtu.edu.cn School of Media and Communication, Shanghai Jiao Tong University, No. 800 Dongchuan Road, Minhang District, Shanghai, 200240, China
*These authors contributed equally to this work.

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users' perceptions of the technology, encompassing perceived usefulness, ease of use, trust, and risk, shape their attitudes (Davis, 1989; Vimalkumar et al., 2021). Thirdly, external factors such as marketplace dynamics, privacy concerns, and ethical considerations can either facilitate or hinder AI adoption (Klaus & Zaichkowsky, 2022; Montes & Goertzel, 2019; Sharma et al., 2022). Finally, individual characteristics like demographics, personality traits, innovativeness, and prior experience predict AI adoption (Liu & Tao, 2022; Venkatesh et al., 2003).

The Technology Acceptance Model (TAM) proposed by Davis et al. (1989) is a widely recognized and extensively used theory for studying user information behavior. It has proven to be effective in explaining user acceptance of new technologies and devices. Previous studies employing TAM have successfully predicted user acceptance of AI products such as service robots (Etemad-Sajadi & Sturman, 2022), medical robots (Liu et al., 2022), and social robots (De Graaf & Allouch, 2013). However, there is a lack of research applying TAM specifically to ChatGPT. Considering that ChatGPT and related technologies have been identified as disruptive innovations (Haque et al., 2022), this study adopts TAM as the theoretical foundation for our model. The reason for choosing the Technology Acceptance Model over other possible models is that it has been reliably scientifically validated in understanding user behavior towards emerging technologies and is a very classical theoretical model, making it a suitable theoretical perspective for understanding user behavior in acceptance. By doing so, we aim to expand the application of TAM theory and gain a better understanding of the factors influencing people's adoption and use of ChatGPT.

To the best of our knowledge, this study serves as the first empirical investigation that employs the TAM model to explore users' intentions and behaviors towards ChatGPT usage. Additionally, we extend the existing TAM framework by considering the moderating effects of demographic variables, namely, gender, geographical area, and educational background. This enables us to examine whether users' intentions to adopt ChatGPT and their usage behavior are influenced by these demographic variables. Ultimately, our aim is to develop theoretical models that have explanatory power for users' intentions to use ChatGPT and their usage behavior and to make theoretical and practical contributions to AI technology adoption research by tailoring the findings to specific user groups. Understanding user adoption of ChatGPT serves as a pivotal cornerstone not only for comprehending its immediate impact but also for informing broader AI integration strategies across various sectors. Insights garnered from this study could offer valuable guidance for businesses, healthcare systems, customer service industries, and beyond, enhancing their approaches to AI implementation. By deciphering the factors influencing ChatGPT adoption, this research contributes to the wider discourse on effective AI utilization, potentially shaping more informed decision-making in diverse fields reliant on AI technologies.

2. Theoretical model and hypothesis development

2.1. Technology adoption model

The TAM, initially proposed by Davis (1989), integrates fundamental constructs from the Theory of Planned Behavior (Ajzen, 1991), including attitude, subjective norms, and perceived behavioral control. TAM has gained significant prominence in the field of information systems as a theoretical framework for understanding individuals' acceptance and adoption of new technologies. Its applicability and predictive power have been extensively validated through numerous empirical studies, supporting its robustness in explaining technology adoption (Mailizar et al., 2021; Marangunic & Granic, 2015; Scherer et al., 2019).

TAM is considered an influential and widely used model in the field of information systems. It has shown good applicability in various domains, including information and computer technologies such as email (Serenko, 2008), mobile banking (Shaikh & Karjaluoto, 2015), and robotics (Pillai et al., 2023). For example, a study investigated students' adoption intentions and actual usage of artificial intelligence-based teacher robots using the TAM (Pillai et al., 2023). They found that perceived ease of use and perceived usefulness played crucial roles among the factors influencing the intention to adopt teacher bots. Moreover, the TAM model has been validated across different countries and cultures, including its applicability in developing countries like China (Liu et al., 2019). The influence of cultural factors on ChatGPT's use environment is crucial. Cultural nuances significantly impact users' perceptions, behaviors, and attitudes towards AI technologies like ChatGPT. For instance, preferences in communication styles, language variations, and social norms can profoundly affect user interactions with ChatGPT (Kuang et al., 2023; Stahl & Eke, 2024). This study specifically examines the adoption and usage patterns of Chinese users, applying the TAM model to understand ChatGPT chatbot usage within the Chinese context. The TAM posits that a person's behavioral intention to use information technology is primarily determined by two key factors: perceived usefulness and perceived ease of use (Davis, 1989). In this study, we identify and retain four core variables of TAM: perceived usefulness, perceived ease of use, behavioral intention, and use behavior.

2.1.1. Perceived usefulness

In the TAM, perceived usefulness refers to the belief that using a specific system will enhance one's job performance and have a positive impact on the user's behavioral intention to adopt the new technology (Davis, 1989; Venkatesh & Davis, 2000).

In the field of AI research, several studies have consistently demonstrated the significant role of perceived usefulness in shaping users' intentions to adopt new technologies. For instance, a study examined the intention to use AI chatbots and found that perceived usefulness positively impacted users' willingness to adopt the technology (Pillai & Sivathanu, 2020). Similarly, another study focused on the determinants of public acceptance of AI

for smart healthcare services and identified perceived usefulness as a key factor driving users' utilization of AI in healthcare contexts (Liu & Tao, 2022). Furthermore, a study investigated users' intentions to use AI voice assistants and found a significant influence of perceived usefulness on users' intentions to use them (Cai et al., 2022). In the realm of customer service, ChatGPT has been integrated into conversational interfaces, providing efficient and personalized support to users, consequently enhancing customer satisfaction and service efficiency (Koc et al., 2023; Oviedo-Trespalacios et al., 2023). In healthcare, ChatGPT's application spans from aiding in medical consultations to streamlining administrative tasks, showcasing its potential in improving patient care and operational workflows (Rahimzadeh et al., 2023).

These studies collectively underscore the importance of perceived usefulness in shaping users' intentions to embrace and engage with AI. Building upon the above findings, the following hypothesis was proposed:

H1. Perceived usefulness would positively influence people's behavioral intention to use ChatGPT.

2.1.2. Perceived ease of use

Perceived ease of use, as conceptualized within the TAM, refers to an individual's perception of the level of effortlessness associated with using a particular system or technology (Davis, 1989). This variable has been consistently linked to users' behavioral intentions to adopt and utilize new technologies. In the domain of AI research, a plethora of studies have underscored the significant role of perceived ease of use in shaping users' intentions to embrace and employ novel AI. For instance, a study demonstrated a positive impact of perceived ease of use on employees' behavioral intention to adopt an AI customer service system within organizations (Chatterjee et al., 2021). Similarly, another recent study explored the factors influencing students' usage of AI robots in educational settings and found that perceived ease of use exerted a significant and positive impact on behavioral intention (Pillai et al.,2023).

A3B2 tlb=.4pt?>Similarly, other studies focusing on various AI technologies have supported this perspective (Liu & Tao, 2022; Pillai & Sivathanu, 2020). In the use of ChatGPT, the visual interface is remarkably intuitive. Furthermore, studies have shown a significant increase in user interaction time and a notable surge in adoption rates post-integration of ChatGPT (Liu & Ma, 2023; Menon & Shilpa, 2023). These observations highlight its ease of use, indicating its effectiveness in enhancing user engagement and facilitating higher adoption rates in practical applications.

Given the positive association between perceived ease of use and behavioral intention established in previous research, we postulate that users will be more inclined to use ChatGPT if they perceive it as easy to use. Therefore, we made the following hypothesis:

H2. Perceived ease of use would positively influence people's behavioral intention to use ChatGPT.

2.1.3. Perceived ease of use and perceived usefulness

According to the TAM, perceived ease of use directly influences perceived usefulness (Davis, 1989). This relationship stems from the notion that the easier a system is to use, the greater the improvement in job performance (Venkatesh & Davis, 2000). Empirical evidence supports this relationship in the context of various AI applications. For instance, in a study of customers' attitudes and behavioral intentions toward robotic restaurant service, researchers found that when customers perceived interactions with service robots as effortless, they also perceived the robots to provide useful and satisfying service, thereby increasing perceived usefulness (Choe et al., 2022). Similarly, this relationship has proven to exist in the field of AI-integrated customer relationship management systems (Chatterjee et al., 2021). Recent studies in the field of artificial intelligence have provided valuable insights into the dynamics of technology adoption. Notably, contemporary research has reiterated the significance of perceived ease of use as a catalyst for enhancing perceived usefulness in AI technologies (Wang et al., 2023). Building upon the discussions regarding the association between perceived ease of use and perceived usefulness, we proposed the following hypothesis:

H3. Perceived ease of use would positively influence the perceived usefulness to use ChatGPT.

2.1.4. Behavioral intention and use behavior

In the TAM, behavioral intention represents an individual's intention to use a specific technology, while use behavior reflects the actual engagement and utilization of that technology (Davis, 1989). In the context of this study, behavioral intention refers to users' intention to utilize ChatGPT, whereas use behavior pertains to their actual engagement with ChatGPT over a period of time. According to the TAM, behavioral intention has a positive influence on use behavior (Davis, 1989). Consistently, numerous studies in the field of AI research have affirmed the relationship between behavioral intention and use behavior. For instance, a study on the adoption of chatbots in consumer travel planning found a positive correlation between behavioral intention and use behavior (Pillai & Sivathanu, 2020). Similarly, a study investigated the acceptance of AI-based digital assistants and demonstrated that consumers' behavioral intentions positively influenced their use of voice-enabled digital assistants (Vimalkumar et al., 2021). Recent investigations have shed light on the crucial link between behavioral intention and actual usage behavior. Their findings emphasize the pivotal role behavioral intent plays in shaping users' real-world engagement with AI technologies (Jo, 2023a, 2023b). Given that behavioral intention is commonly regarded as the closest precursor to actual use behavior in the AI domain, the following hypotheses were proposed:

H4. Behavioral intention would positively influence people's use behavior of ChatGPT.

Meanwhile, extensive research has demonstrated the positive impact of perceived ease of use and perceived usefulness on behavioral intention, which subsequently influences their actual use behavior (Davis, 1989; Venkatesh & Bala, 2008; Venkatesh & Davis, 2000). Meanwhile, combining the above assumptions, perceived ease of use will increase users' perceived usefulness, and with the increase in users' perception of usefulness in ChatGPT usage, users will have a stronger intention to use and thus generate use behavior. Based on these considerations, the following hypotheses were formulated:

H5. Behavioral intention towards ChatGPT would play a mediating role between perceived usefulness and use behavior.

H6. Behavioral intention towards ChatGPT would play a mediating role between perceived ease of use and use behavior.

H7. Perceived usefulness and behavioral intention would play a chain-mediated role between perceived ease of use and use behavior.

2.2. The moderating effects of demographic variables: Gender, geographical area, and educational background

Two core justifications advocate for the incorporation of these demographic characteristics: Firstly, previous research has demonstrated the impact of individual traits on technology adoption (Venkatesh et al., 2003). Secondly, a study highlighted the incorporation of the TAM model within a more comprehensive framework, integrating moderating variables related to individual traits (Legris et al., 2003). Hence, this study seeks to investigate disparities in individual characteristics, specifically emphasizing the moderating effects of gender, geographical area, and educational background.

Firstly, the influence of gender on technology adoption has been extensively investigated in academic research. Notably, one study provided empirical evidence demonstrating that the positive relationship between perceived usefulness and behavioral intention was much stronger among men compared to women (Ong & Lai, 2006). This finding suggests that men are more inclined to consider the utility and productivity-related aspects of new technologies, emphasizing the importance of useful content in attracting male users to use AI. Furthermore, another study investigating the public acceptance of AI for smart healthcare revealed a stronger relationship between perceived ease of use and perceived usefulness among men as compared to women (Liu & Tao, 2022). This finding implies that men may place greater emphasis on the perceived ease of use of technology as a crucial factor in evaluating the usefulness of AI technologies. Recent research findings indicate that male users demonstrate a higher frequency of ChatGPT usage compared to female users, potentially attributed to the inclination of males to incorporate technological tools into their daily tasks (Jo & Park, 2023). However, contrary to this trend, other studies have revealed a positive correlation between gender and ChatGPT practice. Research conducted by Nikita Bodani et al. (2023) suggests that female users tend to exhibit higher ChatGPT practice values in comparison to their male counterparts. Given the notable impact of gender on technology adoption, it becomes imperative to explore whether gender acts as a moderating factor in the utilization of ChatGPT. By considering the interplay between gender and the factors that influence technology adoption, researchers can acquire valuable insights into how ChatGPT can effectively cater to the diverse needs and preferences of different gender groups.

Secondly, the geographical area, particularly the distinction between urban and rural environments, can exert a significant influence on the behavioral intention to adopt technology (Porto et al., 2019; Tandon et al., 2019). For instance, previous research has demonstrated the moderating role of geographic area in the relationship between perceived usefulness, perceived ease of use, and users' behavioral intention to embrace novel technologies (Tandon et al., 2019). Considering the prevailing developmental disparities between urban and rural areas in contemporary China, as evidenced by prior studies, where urban areas tend to be more developed compared to their rural counterparts, it is highly likely to result in disparities in the willingness and behavioral patterns of urban and rural users when it comes to adopting ChatGPT (Yang, 2023). Therefore, it is crucial for researchers to meticulously consider these regional disparities and the relevant contextual factors when examining the adoption process of ChatGPT. These research endeavors can facilitate the customization of technology adoption strategies tailored to the distinct needs of urban and rural user groups, thereby promoting a more inclusive and equitable technology adoption process.

Moreover, a substantial body of prior research has consistently demonstrated the influential role of users' educational backgrounds in shaping the adoption of new technologies (Nguyen et al., 2014; Yu & Liu, 2022). Specifically, empirical studies have indicated that individuals with higher educational attainment are more inclined to try new technologies, integrate them into their daily lives, and demonstrate higher levels of behavioral intention to adopt them compared to those with lower educational backgrounds (Riddell & Song, 2017). Building upon these findings and considering the relationship between behavioral intention and use behavior, it is reasonable to infer that the positive association between behavioral intention and use behavior is more pronounced among individuals with higher levels of education, as they are more likely to possess the necessary skills, knowledge, and confidence to effectively engage with and utilize technology. In contrast, individuals with lower educational backgrounds may face certain barriers or challenges in understanding and effectively using new technologies, resulting in a weaker link between their behavioral intention and actual use behavior.

However, it is crucial to acknowledge that the investigation of how educational background moderates the relationship between perceptual factors and use behavior has not been extensively addressed in the existing literature specifically focused on AI. Therefore, there is an urgent need for further investigation. Such research efforts can provide

valuable insights into the intricate dynamics between educational background and technology adoption, paving the way for tailored strategies and interventions to promote more effective and inclusive technology utilization across different educational contexts. Emerging research has shed light on the significant moderating impact of educational background on the relationship between ChatGPT and customer experience. Studies, such as that by Abdelkader (2023), indicate a notable correlation where participants with higher education background demonstrate a considerably higher average customer experience score with ChatGPT compared to those with lower educational background. This suggests that individuals with higher educational attainment derive greater benefits in terms of customer experience from ChatGPT's functionalities.

In light of these considerations, an exploratory research question arises regarding the moderating role of demographic factors were proposed:

RQ1. Would gender, geographical area, and educational background moderate the aforementioned relationships?

Specifically, the inquiry aimed to investigate whether these demographic variables exerted a significant moderating role in the aforementioned relationships, thereby contributing to a more comprehensive understanding of the dynamics of ChatGPT adoption and usage. The proposed research model is depicted in Figure 1.

3. Methods

3.1. Sample and data collection

In this study, a questionnaire was employed as the primary data collection instrument. We employed an online survey link shared across platforms like WeChat, Xiaohongshu, Douban, Baidu Tieba, Zhihu, and Weibo to reach our predominantly young and middle-aged ChatGPT user base. This ensured visibility within relevant ChatGPT user groups and AI tech forums, encouraging voluntary participation. Convenience sampling, chosen for its practicality and ease of access, was the selected sampling method. This method was adopted considering the limitations in access to a more comprehensive sampling framework due to time and resource constraints. Data collection lasted one week.

At the outset of the questionnaire, participants were explicitly informed about the anonymous nature of the survey and that their responses would be solely used for academic purposes. Prior to proceeding, all respondents provided their informed consent. To incentivize participation, a compensation of 5 RMB was offered for each completed questionnaire meeting the specified criteria. The initial question of the questionnaire served as a screening mechanism, as it inquired whether the respondent had used ChatGPT. Out of 1,242 distributed questionnaires, 949 respondents replied positively, with 784 deemed valid, resulting in an 82.61% validity rate. During validation, responses completing the survey within 120 seconds were excluded, often indicating rushed or inattentive completion, a standard criterion supported by numerous prior studies. Additionally, two general knowledge questions, like identifying the direction of the sunrise, were included to filter out respondents failing attention checks and ensure data reliability.

The 784 participants' demographics are outlined in Table 1, affirming the sample's representativeness. Participants comprised approximately 58.93% males and 41.07% females. The majority, 69.26%, fell within the 19–25 age range, while 68.88% held bachelor's degrees. Geographically, 71.81% were urban residents, with 28.19% residing in rural areas.

3.2. Measurement

The questionnaire measured two TAM core variables, perceived usefulness and perceived ease of use, as well as two outcome variables, behavioral intention and use behavior.

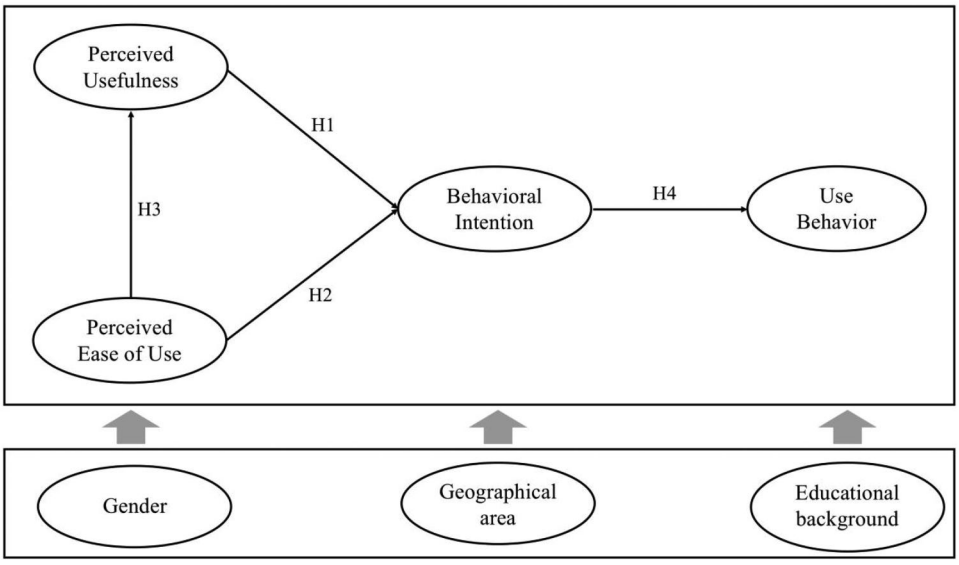


Figure 1. The proposed research model.
Note. PU = Perceived usefulness; PEU = Perceived ease of use; BI = Behavioral intention; USE = Use behavior. Factor loading are standardized.

The measurement items were adapted from existing scales, all of which were on a 5-point Likert scale, except that the use behavior was multiple choices. The purpose of the adaptation was to understand users' perceptions and usage behaviors of ChatGPT, which prompted us to adapt an existing scale originally designed for web-based systems by modifying the topic research object in the original questionnaire to a question-asking question about ChatGPT usage. As presented in Table 2, items for each variable showed good reliability, with every Cronbach's α greater than 0.75.

3.3. Data analyses

To examine the research hypotheses and questions, the current study employed structural equation modeling (SEM). One of the advantages of SEM for analysis is that it allows latent variables, which can help effectively control measurement errors, as well as precisely estimate mediating and moderating effect. SEM makes the validation of TAM model more flexible and accurate.

Data analysis was conducted using Mplus software (version 8.3). In line with the recommendations of Kline (2011), several criteria were considered during the SEM analysis, which jointly showed the degree of fitting and testing power of the existing model. These criteria included comparative fit index (CFI) and Tucker-Lewis index (TLI) values above 0.90, a confirmatory fit index to degrees of freedom ratio (CMID/DF) less than 5, and a root mean square error of approximation (RMSEA) below 0.08. Meeting the above requirements of indicators is the basis for the series of subsequent analyses. Following the assessment of convergent validity and discriminant validity, the significance of direct and indirect effect paths between the key variables was analyzed using the partial least squares (PLS) method. Subsequently, the moderating roles of gender, geographical area, and educational background were examined through multigroup path analyses. Initially, all model paths were estimated separately for the two participant groups. Each path was then constrained to be equal between the groups to test for potential differences in model coefficients across the moderators.

Table 1. Demographic indicators of participants.

	N (%)	
Characteristics	N	%
Gender		
Male	462	58.93
Female	322	41.07
Geographical area		
Urban	563	71.81
Rural	221	28.19
Educational background		
Senior high school and below	12	1.53
Junior college	35	4.46
Bachelor degree	540	68.88
Master degree	147	18.75
PhD degree	50	6.38
Age		
<18	20	2.55
19–25	543	69.26
25–30	185	23.60
30–40	35	4.46
>40	1	0.13
Social role		
Student	556	70.92
Working person	228	29.08

Note. N = 784.

4. Results

4.1. Convergent validity

As depicted in Table 3, the factor loadings for all items surpassed 0.60, with the majority exceeding 0.70. Additionally, all composite reliability (CR) values exceeded the threshold of 0.70, with the lowest value being 0.76. Though a few average variance extracted (AVE) values fell slightly below the recommended threshold of 0.50, their associated CR values exceeded 0.6 by a considerable margin. Referring to previous research, AVE can be accepted if it is higher than 0.4 when CR is higher than 0.6, so the convergent validity of the construct is still adequate (Fornell & Larcker, 1981; Lam, 2012). To sum up, in accordance with the guidelines outlined by Fornell and Larcker (1981) and Hair et al. (2016), these three criteria collectively provide sufficient evidence supporting the convergent validity of the constructs under investigation; that is, the measurements are relatively good to get the concepts that we are studying.

4.2. Discriminant validity

According to Fornell and Larcker (1981), the square roots of AVE and correlations could be adopted to test discriminant validity, referring to the extent to which a specific variable was not related to others that measure different constructs. As shown in Table 4, the square root of the AVE of each variable displayed in the diagonal was higher than the correlations in the row and column that correspond to that construct, establishing great discriminant validity among

Table 2. Measures and reliabilities of main variables.

Variable	Number of items	Type of items	Sample item	Cronbach's α	Reference
PU	4	5-point Likert scale	Using ChatGPT help improve my academic or work performance.	0.852871	Venkatesh and Bala (2008)
PEU	4	5-point Likert scale	I found ChatGPT very easy to use.	0.786351	Venkatesh and Bala (2008)
BI	4	5-point Likert scale	I will be using ChatGPT a lot in the future.	0.778985	Venkatesh and Bala (2008); Moon and Kim (2001)
USE	3	Multiple choice	On average, how many hours per week do you use ChatGPT? (1 h and less /1–2 h /3–6 h/6–8 h/more than 8 h)	0.756607	Moon and Kim (2001)

Note. PU = Perceived usefulness; PEU = Perceived ease of use; BI = Behavioral intention; USE = Use behavior.

Table 3. Construct indicators of convergent validity.

Variables	Items	Significance estimate			Item reliability		Composite reliability	Convergent validity
		Unstandardized estimate	S.E.	*p* value	Item loading	SMC	Standardized Estimate (CR)	AVE
PU	PU1	1			0.771	0.594	0.853	0.593
	PU2	1.016	0.047	<0.001	0.781	0.610		
	PU3	1.03	0.047	<0.001	0.784	0.615		
	PU4	1.013	0.049	<0.001	0.743	0.552		
PEU	PEU1	1			0.743	0.552	0.787	0.481
	PEU2	0.928	0.058	<0.001	0.63	0.397		
	PEU3	1.035	0.058	<0.001	0.707	0.500		
	PEU4	1.04	0.06	<0.001	0.689	0.475		
BI	BI1	1			0.651	0.424	0.784	0.478
	BI2	1.311	0.087	<0.001	0.668	0.446		
	BI3	1.404	0.084	<0.001	0.792	0.627		
	BI4	1.139	0.078	<0.001	0.644	0.415		
USE	USE1	1			0.706	0.498	0.759	0.513
	USE2	0.977	0.063	<0.001	0.685	0.469		
	USE3	1.118	0.069	<0.001	0.756	0.572		

Note. PU = Perceived usefulness; PEU = Perceived ease of use; BI = Behavioral intention; USE = Use behavior. N = 784.

Table 4. Construct indicators of discriminant validity.

	Convergent validity	Discriminant validity			
	AVE	PU	PEU	BI	USE
PU	0.593	**0.770**			
PEU	0.481	0.768	**0.693**		
BI	0.478	0.453	0.457	**0.691**	
USE	0.513	0.322	0.414	0.624	**0.716**

Note. PU = Perceived usefulness; PEU = Perceived ease of use; BI = Behavioral intention; USE = Use behavior. N = 784.

variables, that is, there are significant differences between the different concepts studied.

4.3. Structural equation model

Based on the research hypotheses, a structural equation model (SEM) was constructed, as illustrated in Figure 2. The goodness-of-fit indicators for this model are presented in Table 5, indicating a favorable fit.

The direct path analysis results in the model are shown in Table 6. The effects of all paths were significant, including "perceived usefulness → behavioral intention" ($\beta = 0.227$, $p < 0.01$), "perceived ease of use → behavioral intention" ($\beta = 0.304$, $p < 0.001$), "perceived ease of use → perceived usefulness" ($\beta = 0.767$, $p < 0.001$), and "behavioral intention → use behavior" ($\beta = 0.639$, $p < 0.001$), confirming Hypotheses 1–4.

As presented in Table 7, further indirect path analysis results showed that behavioral intention mediated the relation between perceived usefulness and use behavior ($\beta = 0.145$, $p = 0.015$), and behavioral intention mediated the relation between perceived ease of use and use behavior ($\beta = 0.195$, $p = 0.001$), while the sequential pathway "perceived ease of use → perceived usefulness → behavioral intention → use behavior" was significant ($\beta = 0.111$, $p = 0.015$), meaning that perceived usefulness and behavioral intention could mediate the relationship between perceived ease of use and use behavior. Hypotheses 5–7, which indicated the mediation roles of variables, are confirmed.

4.4. Moderating effect

The results of the moderation analysis, presented in Table 8, provide insights into the influence of gender, geographic area, and educational background as grouping variables and it is an exploration of Research Question 1 about moderating effects. To provide clarity, the demographic breakdown employed in the study encompasses the following classifications: male/female for gender, city/countryside for geographic area, and undergraduate or below/postgraduate or above for educational background. Notably, the effect of behavioral intention on use behavior was found to be significantly moderated by gender (*coefficient difference* = 0.109, $p < 0.001$), geographic area (*coefficient difference* = 0.066, $p = 0.045$), and educational background (*coefficient difference* = 0.036, $p < 0.001$). These findings suggest that the relationship between behavioral intention and use behavior varies based on gender, geographic area, and educational background. Furthermore, educational background exhibited a moderating effect on the relationship between perceived ease of use and perceived usefulness (*coefficient difference* = 0.126, $p = 0.001$), as well as between perceived ease of use and behavioral intention (*coefficient difference* = 0.030, $p = 0.038$). This implies that the impact of perceived ease of use on both perceived usefulness and behavioral intention differs depending on the participants' educational backgrounds. However, the remaining moderating effects on specific paths did not yield statistically significant results, indicating that gender, geographic area, and educational background did not significantly moderate those particular relationships in the model.

5. Discussion

Presently, ChatGPT has undergone remarkable advancements and gained widespread popularity as an AI model. Nevertheless, there is a dearth of empirical studies delving into the factors and influencing mechanisms that drive users' willingness to adopt it. This study employs the TAM as the theoretical framework to analyze users' intentions and

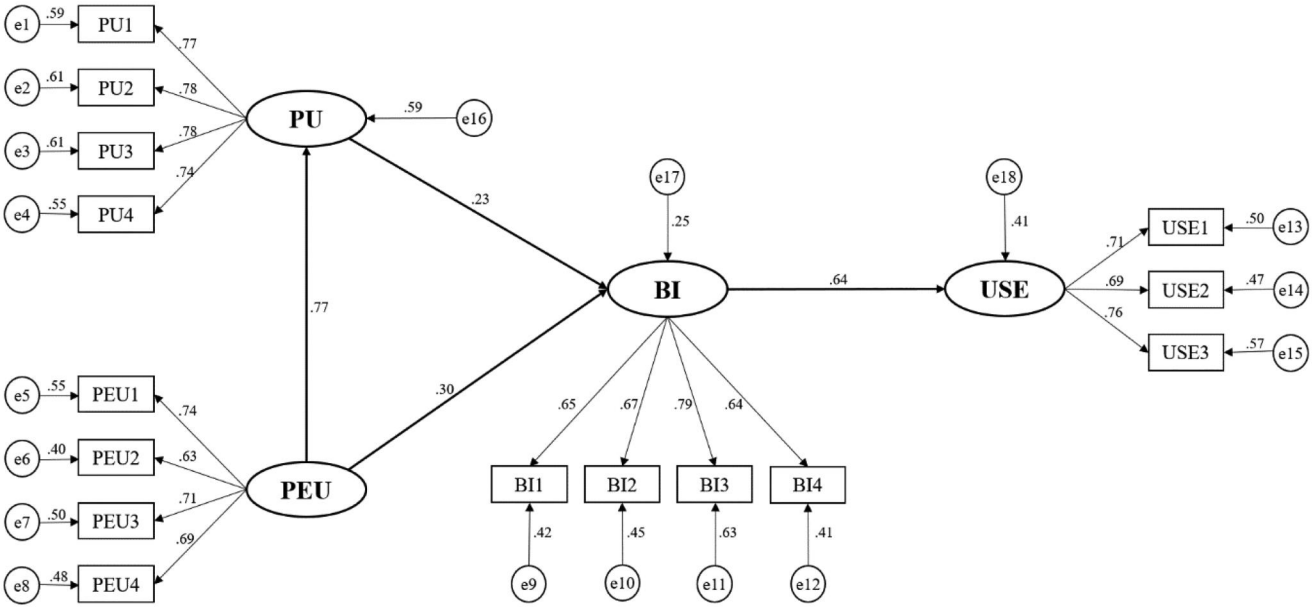


Figure 2. The demonstration of the structural model.

Table 5. The goodness-of-fit indicators of SEM.

Indicator	Value
CMID	201.363
DF	86
CMID/DF	2.341
GFI	0.966
AGFI	0.952
CFI	0.974
TLI (NNFI)	0.968
RMSEA	0.041
SRMR	0.0358

Table 6. PLS-SEM path coefficients.

Path			Unstandardized estimate	S.E.	*p* value	CR (factor loading)	R^2
PU	→	BI	0.244	0.081	0.003	0.227	0.250
PEU	→	BI	0.346	0.089	<0.001	0.304	
PEU	→	PU	0.814	0.053	<0.001	0.767	0.589
BI	→	USE	0.782	0.062	<0.001	0.639	0.409

Note. PU = Perceived usefulness; PEU = Perceived ease of use; BI = Behavioral intention; USE = Use behavior. N = 784.

Table 7. PLS-SEM indirect effects path coefficients.

	Std. estimate	S.E.	EST./S.E.	*p* value
Effects from PU to USE				
Total indirect	0.145	0.06	2.427	0.015
Specific indirect path				
PU→BI→USE	0.145	0.06	2.427	0.015
Effects from PEU to USE				
Total indirect	0.306	0.036	8.407	<0.001
Specific indirect path 1				
PEU→BI→USE	0.195	0.057	3.44	0.001
Specific indirect path 2				
PEU→PU→BI→USE	0.111	0.046	0.422	0.015

Note. PU = Perceived usefulness; PEU = Perceived ease of use; BI = Behavioral intention; USE = Use behavior. N = 784.

behaviors in the context of ChatGPT adoption. Furthermore, the study looked at the moderating impact of demographic characteristics such as gender, geographic area, and educational background. The acquired data supported all of the specified study hypotheses (H1–H7). It was revealed that both perceived usefulness and perceived ease of use had a positive influence on behavioral intention (H1, H2). Furthermore, it was discovered that perceived ease of use positively influenced perceived usefulness (H3). Behavioral intention positively influenced usage behavior (H4). Behavioral intention acted as a mediating factor in the relationship between perceived usefulness and use behavior, as well as the relationship between perceived ease of use and use behavior (H5, H6). Perceived usefulness and behavioral intention played a chain-mediated role between perceived ease of use and use behavior (H7).

Gender was found to act as a moderating variable, indicating that the relationship between behavioral intention and use behavior exhibited greater strength among females compared to males. Similarly, geographic area served as a moderating factor, highlighting that the association between behavioral intention and use behavior was found to be stronger among urban users in comparison to their rural counterparts. Moreover, educational background played a crucial role as a moderating variable. This study revealed that the connections between perceived ease of use and perceived usefulness, perceived ease of use and behavioral intention, and behavioral intention and use behavior were observed to be stronger among individuals with higher educational backgrounds relative to those with lower educational backgrounds.

5.1. Correlations among user perceptions and behaviors

Within the context of the widespread utilization of AI, it is imperative to conduct studies that empirically measure the impact of perceived usefulness and perceived ease of use on behavioral intention and use behavior towards new technological advancements. Firstly, the findings of this study demonstrate that perceived usefulness exerts a statistically significant and positive influence on users' behavioral

Table 8. Multigroup analyses with gender, geographical area and educational background as moderators.

Paths			Moderator	Path coefficient		Group 2 – Group 1	
				Group 1	Group 2	Coefficient difference	*p* value
				Male	Female		
PEU	→	PU	Gender	0.101***	0.132***	0.031	0.464
PU	→	BI		0.032*	0.06**	0.028	0.179
PEU	→	BI		0.038**	0.06**	0.022	0.092
BI	→	USE		0.147***	0.256***	**0.109**	**<0.001**
				Rural	Urban		
PEU	→	PU	Geographic	0.116***	0.121***	0.005	0.898
PU	→	BI	area	0.062**	0.048***	−0.014	0.504
PEU	→	BI		0.068**	0.030*	−0.039	0.092
BI	→	USE		0.151***	0.218***	**0.066**	**0.045**
				Undergraduate or below	Postgraduate or above		
PEU	→	PU	Educational	0.084***	0.211***	**0.126**	**0.001**
PU	→	BI	background	0.041*	0.070*	0.028	0.201
PEU	→	BI		0.032*	0.062**	**0.030**	**0.038**
BI	→	USE		0.172***	0.268***	**0.036**	**<0.001**

Note. PU = Perceived usefulness; PEU = Perceived ease of use; BI = Behavioral intention; USE = Use behavior. N = 784.

intentions regarding the adoption of ChatGPT. This outcome aligns with previous studies (Cascella et al., 2023; Davis, 1989; Venkatesh & Davis, 2000), thus further substantiating the notion that perceived usefulness represents a pivotal factor that influences behavioral intention. This result can be comprehended in terms of the functional aspects of AI. As an emerging AI technology, ChatGPT interacts with users through textual dialogue, yet its functionality extends far beyond initial expectations. In practical applications, ChatGPT is capable of generating diverse forms of content, encompassing news reports, advertising copy, and even novels. Accordingly, the more users perceive ChatGPT to be useful, the more likely they would use it.

Secondly, the findings of this study reveal a significant positive correlation between users' perceived ease of use and their behavioral intentions towards ChatGPT. This outcome aligns with the fundamental theoretical framework of the TAM originally proposed by Davis (1989) and is consistent with prior empirical research in related domains (Ashfaq et al., 2020; Venkatesh & Davis, 2000). One plausible explanation stems from the intuitive and user-friendly interface design of ChatGPT. The interface is deliberately crafted to be straightforward, allowing users to swiftly engage with the system by simply entering their desired content into a dialog box to promptly obtain the information they seek. This seamless and effortless interaction enables users to easily grasp the functional capabilities of ChatGPT and initiate conversations to accomplish their tasks effectively. Consequently, users' behavioral intentions to adopt ChatGPT are significantly bolstered when they perceive that their desired goals can be readily achieved through the utilization of ChatGPT.

Thirdly, the findings from the path analysis demonstrate a significant positive relationship between perceived ease of use and perceived usefulness in the context of Chinese users' adoption of ChatGPT. This observation aligns with prior research (Choe et al., 2022; Davis, 1989; Venkatesh & Davis, 2000) that suggests users' perceptions of the system's ease of use directly influence their evaluations of its overall usefulness. Specifically, when users perceive ChatGPT as being easy to use, it enhances their perception of the system's practical value and effectiveness in facilitating their specific tasks and objectives. Consequently, this finding underscores the critical importance of designing systems that are perceived as user-friendly and easy to navigate, as these design characteristics play a pivotal role in shaping users' perceptions of usefulness.

Fourthly, within the context of ChatGPT utilization, behavioral intention demonstrates a positive influence on actual usage behavior. This discovery aligns with the foundational principles of the Technology Acceptance Model (TAM) and is substantiated by prior empirical investigations (Alzahrani et al., 2019; Davis et al., 1989; Moon & Kim, 2001; Pillai & Sivathanu, 2020), validating behavioral intention as a direct precursor to usage behavior. A study by Pillai and Sivathanu (2020) provided evidence that users' intention to engage with AI-powered intelligent chat systems positively impacts their actual usage behavior. This finding implies that bolstering users' intention to adopt ChatGPT represents the most straightforward and effective approach to fostering real usage behavior.

5.2. Intermediary effects

Another noteworthy contribution of our study is the thorough investigation into the mediating role of behavioral intention in the intricate relationship between perceived factors, namely perceived usefulness and perceived ease of use, and use behavior. Firstly, our findings provide compelling evidence that behavioral intention towards ChatGPT functions as a mediator between perceived usefulness and use behavior. This implies that users' assessment of the system's usefulness not only directly influences their behavioral intention but also indirectly impacts their use behavior through the mediation of behavioral intention (Venkatesh & Davis, 2000). This discovery accentuates the influential power of users' perceptions regarding the system's usefulness in shaping their behavioral decisions (Cai et al., 2022; Davis, 1989; Pillai & Sivathanu, 2020), thereby offering invaluable

guidance to system designers in their efforts to enhance system effectiveness and foster user engagement.

Secondly, our investigation has uncovered that users' behavioral intention towards ChatGPT plays a mediating role between perceived ease of use and use behavior. In simpler terms, users' perception of ChatGPT's ease of use augments their behavioral intention, ultimately increasing the likelihood of engaging in use behavior. This finding underscores the paramount importance of designing interfaces and interactions that are user-friendly, promoting enhanced perceived ease of use (Liu & Tao, 2022; Pillai et al.,2023). By optimizing the system's ease of use, designers can effectively heighten users' behavioral intentions, thereby fostering a more active adoption of the ChatGPT system (Degen & Ntoa, 2022).

Furthermore, our study has revealed that users' perceived usefulness of ChatGPT assumes a mediating role between perceived ease of use and behavioral intention. This implies that users' perception of the system's ease of use not only directly influences their perceived usefulness but also indirectly affects their behavioral intentions through their perception of usefulness (Davis, 1989; Venkatesh & Davis, 2000; Vimalkumar et al., 2021).

5.3. The moderating effects of demographic variables

In addition to empirically testing our hypothesized model, this study raises an important exploratory research question regarding the potential moderating role of demographic variables in the relationships under investigation.

5.3.1. The moderating role of gender

First, we found that gender moderated the positive relationship between behavioral intention and use behavior. Notably, the relationship between behavioral intention and use behavior exhibited greater strength among females compared to males. To explain this finding, we draw upon the Theory of Planned Behavior (Ajzen, 1991), which suggests that behaviors that are not fully controlled by individual will are not only influenced by behavioral intentions but also by the actual control conditions such as the individual's ability, opportunity, and resources to perform the behavior. Prior research has suggested gender disparities in computer skills and information technology proficiency (Litt, 2013; Taylor & Dalal, 2017), indicating that women may face greater challenges when engaging with AI technologies. This proficiency gap might incentivize women to actively seek out and engage with AI tools like ChatGPT to bridge this technological divide. Therefore, it is plausible that women, driven by the motivation to enhance their proficiency, exhibit more proactive behavior in utilizing ChatGPT, thereby impacting the relationships observed in this study.

Second, we did not observe a significant gender difference in the relationship between perceived usefulness and behavioral intention. Despite previous studies reporting a stronger association between perceived usefulness and behavioral intention among men compared to women (Ong & Lai, 2006; Terzis & Economides, 2011), our findings did not support such a discrepancy. We speculate that this divergence might be attributed to the focus of previous studies on investigating the use behavior of e-learning tools such as the Internet and computers, while our study specifically targeted ChatGPT as a novel AI. Another potential explanation lies in the extraordinary capabilities of ChatGPT, which can comprehend users' intentions, emotions, and knowledge to provide answers and responses accordingly. Consequently, irrespective of gender, once users perceive ChatGPT as useful, their willingness to utilize it becomes relatively robust. Remarkably, a study on AI reported that the gender moderating effect was not significant (Belanche et al., 2019), suggesting the need for further validation with more data in future studies.

Third, our study revealed no significant gender differences in the impact of perceived ease of use on behavioral intention, which diverges from prior research. Previous studies have indicated that the relationship between perceived ease of use and behavioral intention tends to be stronger in women compared to men (Ong & Lai, 2006; Terzis & Economides, 2011). Nevertheless, our investigation with ChatGPT users showed that gender did not significantly moderate the effect of perceived ease of use on behavioral intention. This unexpected finding may be attributed to the simplicity of ChatGPT's design and functional features (Malik et al., 2023; Shoufan, 2023). As an AI model, ChatGPT's interaction heavily relies on language and comprehension skills, which may not be strongly associated with gender. Consequently, ChatGPT might exhibit minimal gender-based differences in users' ease of use requirements, resulting in the absence of a significant gender effect on the relationship between perceived ease of use and behavioral intention. It is imperative to acknowledge that this finding does not provide a comprehensive conclusion on the influence of gender on AI product acceptance and usage. Future research could adopt a more comprehensive theoretical framework and utilize multidimensional data analysis to thoroughly explore the potential role of gender in AI product acceptance and adoption.

Fourth, we did not observe a significant gender difference in the relationship between perceived ease of use and perceived usefulness, which aligns with previous research findings (Forgas-Coll et al., 2022; Ong & Lai, 2006). We propose that this congruence may be attributed to ChatGPT's adherence to the psychological contract in its design, wherein interactions with users are thoughtfully crafted to align with their expectations and habits. For instance, ChatGPT strives to provide concise and easily comprehensible responses to minimize users' cognitive load. This design approach aligns with the psychological contract, ultimately heightening users' perception of ChatGPT's usefulness. Thus, when encountering the novel technology of ChatGPT, regardless of gender, users perceive its ease of use, consequently attributing usefulness to the technology.

5.3.2. The moderating role of geographical area

From a geographical area perspective, the study found that geographical area moderated the relationship between

behavioral intention and use behavior. The findings suggest that the association between behavioral intention and use behavior is stronger for urban users compared to rural users. Three possible explanations can account for this phenomenon. Firstly, notable differences exist between urban and rural users concerning their education level, exposure to technology, and digital literacy (Freeman et al., 2020; Yang & Zhang, 2020). Urban areas typically possess abundant educational resources and access to technological information (Behrend et al., 2011), potentially facilitating urban users' comprehension and utilization of AI technologies like ChatGPT. Secondly, urban users frequently encounter more intricate and diverse life needs and information interaction scenarios (Jamei et al., 2017). As ChatGPT operates as a language-driven AI model with heightened requirements and comprehension of user inputs, and considering that urban users are regularly exposed to a plethora of complex information exchange and problem-solving scenarios in their daily lives (Quan, 2022), this heightened exposure may intensify their demand for utilizing ChatGPT. Consequently, this reinforcement can strengthen the linkage between behavioral intention and use behavior. Additionally, urban and rural users differ in their acceptance of digital technology and sociocultural factors (Lythreatis et al., 2022; Marshall et al., 2022). Urban users may exhibit more openness and receptivity towards emerging technologies, whereas rural users may lean towards a conservative stance regarding new technologies (Haggstrom et al., 2019). Such disparity in attitude could potentially lead to greater engagement and active use of ChatGPT among urban users, consequently enhancing the connection between their behavioral intention and use behavior.

Interestingly, this study did not find a moderating effect of the urban-rural divide in the relationships between perceived ease of use and perceived usefulness, between perceived ease of use and behavioral intention, or between perceived usefulness and behavioral intention. One possible explanation for this finding is that, in the context of ChatGPT usage, the importance of urban and rural distinctions does not significantly influence these relationships, or at the very least, the observed differences are not substantial. However, it is important to note that these hypotheses warrant further investigation to attain a comprehensive understanding of the nuanced dynamics at play.

5.3.3. The moderating role of educational background

It was found that educational background played a crucial role as a moderating variable. Firstly, the positive relationship between perceived ease of use and perceived usefulness is more pronounced for highly educated users relative to those with lower educational backgrounds. This finding is in line with previous research demonstrating that individuals with a higher background in education are more inclined to utilize information technology for work-related purposes (Riddell & Song, 2017; Yu & Liu, 2022) and engage in activities that enhance their intellectual capital, such as seeking professional information services (Hargittai & Hinnant, 2008). As ChatGPT is closely tied to these online activities (Cascella et al., 2023; Malik et al., 2023) and exhibits competence and efficiency in accomplishing related tasks, it proves especially beneficial for the highly educated population. Consequently, users with a higher educational background are more likely to perceive ChatGPT as a valuable tool when it is perceived to be easy to use.

Secondly, the connections between perceived ease of use and behavioral intention were observed to be stronger among individuals with higher educational backgrounds relative to those with lower educational backgrounds. This novel finding can be explained by considering the functional role of AI technology for users. Previous studies have shown that individuals with a higher educational background are more inclined to view IT as a tool for utilitarian purposes than for entertainment (Bonfadelli, 2002; Shah & Kwak, 2001). Therefore, compared to individuals with a lower educational background, those with a higher educational background are more likely to use ChatGPT extensively in their daily work. Consequently, these users prioritize the ease of use of ChatGPT as a practical tool for their daily tasks. Hence, when users with a high educational background perceive ChatGPT to be easier to use, they are more likely to adopt it for their work-related needs. Furthermore, this study found no moderating effect of educational background on the relationship between perceived usefulness and behavioral intention. This may be attributed to the remarkable features offered by ChatGPT, which lead to users' willingness to utilize it regardless of their educational background as long as they perceive its usefulness.

Thirdly, the observed amplified relationship between behavioral intention and use behavior among individuals with higher educational backgrounds may be linked to their predisposition towards technology. Existing literature has consistently highlighted the influence of educational background on attitudes toward technology adoption (Bonfadelli, 2002; Litt, 2013). Individuals with advanced education often exhibit a stronger inclination towards perceiving new technology as crucial and integral, leading to a more immediate and enthusiastic embrace of innovative tools like ChatGPT if they have a higher level of use intention. Hence, those with higher educational backgrounds tend to demonstrate a more proactive intention toward promptly adopting and integrating ChatGPT into their workflows. This discovery offers a novel perspective in the realm of AI adoption, underscoring the necessity for further comprehensive investigations in subsequent studies.

5.4. Implications

As an emerging technology, ChatGPT has experienced rapid growth in its user base. However, there remain certain users who approach ChatGPT with caution and hesitation (Shoufan, 2023). Consequently, the objective of this study is to advance users' understanding of new technology acceptance behaviors and make a theoretical and practical contribution to the adoption and acceptance of ChatGPT technology by extending the TAM study to developing countries.

Firstly, it is imperative to conduct studies that examine TAM models within diverse technological, user group, and organizational contexts (Hu et al., 1999). While numerous recent studies have applied TAM to AI adoption in developed countries (Cai et al., 2022; Mohr & Kühl, 2021), a noteworthy contribution of this study lies in being the first investigation to apply TAM to measuring user adoption of ChatGPT within developing countries. Consequently, the findings offer insights from users in developing countries that advance the development of ChatGPT.

Secondly, through an investigation of the determinants of ChatGPT users' willingness to adopt and their usage behavior, this study reveals that perceived usefulness and perceived ease of use exert a significant and positive influence on users' adoption intentions and behaviors towards ChatGPT. These insights provide valuable contributions to user research and marketing studies pertaining to AI. Consequently, future dissemination and promotion efforts for ChatGPT should prioritize users' evaluations of the technology's usefulness and ease of use, with the aim of increasing the overall adoption rate of AI.

Lastly, while previous TAM studies have examined demographic differences, particularly in terms of gender (Liu & Tao, 2022; Zhang et al., 2023), fewer studies have emphasized differences in users' geographic areas and educational backgrounds. Another noteworthy theoretical contribution of this study lies in predicting the influential factors contributing to differences in ChatGPT usage based on demographic variables such as gender, geographic area, and educational background. From a practical standpoint, gender differences are clearly elucidated through this study. It underscores the focus of female users on the actual use behavior of new technology, especially the heightened propensity for active usage among women when there is a stronger behavioral intention to use ChatGPT. Consequently, future efforts in promoting and implementing AI should aim to address women's concerns regarding their willingness to use AI, thereby increasing the actual adoption of the technology. In addition to unveiling geographic disparities, this study illuminates that urban users exhibit a more pronounced positive influence of behavioral intention on use behavior. Moreover, it identifies distinctions in the impact of educational backgrounds, indicating that highly educated users are more inclined to engage in actual use behaviors driven by favorable attitudes towards technology. These revelations offer valuable insights for marketers seeking to craft tailored AI adoption strategies that resonate with individuals of varying educational backgrounds. This study underscores the significance of delving into individual differences in future TAM research, emphasizing the need for researchers to incorporate and address these nuanced aspects when developing and validating TAM in subsequent studies.

5.5. Limitations and future recommendations

There are certain areas in the current study that would benefit from improvement. Firstly, the sample's potential lack of representation of ChatGPT users may impact the findings' generalizability. Specifically, the distribution of the questionnaire through specific social networking platforms in China could have limited sample diversity, particularly concerning geographic and cultural representation. There is a need for more diverse sampling strategies and expanded distribution channels in future research to capture a varied user base, enhancing the study's overall applicability and representativeness. Secondly, the current study focused on adopters of ChatGPT during its nascent technological stage. Given the dynamic nature of technology acceptance, behavioral intention and use behavior towards ChatGPT may evolve as the technology undergoes iterative updates and users gain more experience with it. To explore the temporal changes in acceptance and behavior, future longitudinal studies can be conducted to validate and extend the current findings. Thirdly, this study exclusively examined users from developing countries. It is recommended that future research investigate respondents from different countries and conduct cross-cultural comparisons to ascertain potential differences in TAM regarding ChatGPT usage behaviors across various cultures. Such cross-cultural analyses would contribute to the validation and enhancement of TAM, enabling the prediction of future adoption behaviors for emerging ICTs in diverse cultural contexts. Furthermore, while recognizing the robustness of TAM, it is important to acknowledge its limitations, especially in a rapidly evolving field of AI like ChatGPT. While this study did not consider variables such as initial trust, perceived risk, and anthropomorphic traits within the TAM framework (Al-Adwan et al., 2023; Quan et al., 2023), it is critical to acknowledge the importance of these factors in prior research. Future research should extend the TAM by integrating these variables to provide a more comprehensive understanding of user adoption behavior in AI environments, including ChatGPT.

6. Conclusion

This study aimed to explore the factors influencing users' adoption of ChatGPT, with a specific focus on investigating the potential moderating effects of gender, geographical area, and educational background. The findings provide valuable insights into the determinants of use behavior. The results indicated that perceived usefulness, perceived ease of use, behavioral intention, and use behavior were positively correlated with each other. Furthermore, behavioral intention acted as a mediator in the relationship between perceived usefulness and use behavior, as well as between perceived ease of use and use behavior. Perceived usefulness and behavioral intention played a chain-mediated role between perceived ease of use and usage behavior. Of particular significance, the study identified gender, geographical area, and educational background as moderators in certain relationships. Specifically, the relationship between behavioral intention and use behavior exhibited greater strength among females compared to males. Similarly, the association between behavioral intention and use behavior was found to be stronger among urban users in comparison to their rural

counterparts. Moreover, this study revealed that the connections between perceived ease of use and perceived usefulness, perceived ease of use and behavioral intention, and behavioral intention and use behavior were observed to be stronger among individuals with higher educational backgrounds relative to those with lower educational backgrounds. These insights not only enrich the current understanding of AI adoption but also underscore the essential need for customized strategies targeting diverse user demographics. These strategies are pivotal in the design, implementation, and marketing of ChatGPT and similar AI technologies. Limitations include a narrow sample from specific platforms and a focus on early adopters. Future studies should diversify samples, explore cross-cultural differences, and integrate other factors (e.g., risk factors) for a fuller understanding of AI adoption in ChatGPT contexts.

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ORCID

Jiaojiao Ma http://orcid.org/0000-0001-8683-7861
Pengcheng Wang http://orcid.org/0000-0002-0241-1888
Tian Wang http://orcid.org/0009-0003-0676-4650
Xiang Shan Pang http://orcid.org/0009-0005-6777-2149
Dake Wang http://orcid.org/0000-0001-6486-8827

References

Abdelkader, O. A. (2023). ChatGPT's influence on customer experience in digital marketing: Investigating the moderating roles. *Heliyon*, *9*(8), e18770–e18770. https://doi.org/10.1016/j.heliyon.2023.e18770

Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, *50*(2), 179–211. https://doi.org/10.1016/0749-5978(91)90020-T

Al-Adwan, A. S., Li, N., Al-Adwan, A., Abbasi, G. A., Albelbisi, N. A., & Habibi, A. (2023). Extending the technology acceptance model (TAM) to predict university students' intentions to use metaverse-based learning platforms. *Education and Information Technologies*, *28*(11), 1–33. https://doi.org/10.1007/s10639-023-11816-3

Alzahrani, A. I., Mahmud, I., Ramayah, T., Alfarraj, O., & Alalwan, N. (2019). Modelling digital library success using the DeLone and McLean information system success model. *Journal of Librarianship and Information Science*, *51*(2), 291–306. https://doi.org/10.1177/0961000617726123

Ashfaq, M., Yun, J., Yu, S., & Loureiro, S. M. C. (2020). I, Chatbot: Modeling the determinants of users' satisfaction and continuance intention of AI-powered service agents. *Telematics and Informatics*, *54*, 101473. https://doi.org/10.1016/j.tele.2020.101473

Behrend, T. S., Wiebe, E. N., London, J. E., & Johnson, E. C. (2011). Cloud computing adoption and usage in community colleges. *Behaviour & Information Technology*, *30*(2), 231–240. https://doi.org/10.1080/0144929X.2010.489118

Belanche, D., Casaló, L. V., & Flavián, C. (2019). Artificial intelligence in FinTech: Understanding robo-advisors adoption among customers. *Industrial Management & Data Systems*, *119*(7), 1411–1430. https://doi.org/10.1108/IMDS-08-2018-0368

Bodani, N., Lal, A., Maqsood, A., Altamash, S., Ahmed, N., & Heboyan, A. (2023). Knowledge, attitude, and practices of general population toward utilizing ChatGPT: A cross-sectional study. *SAGE Open*, *13*(4), 21582440231211079. https://doi.org/10.1177/21582440231211079

Bonfadelli, H. (2002). The internet and knowledge gaps: A theoretical and empirical investigation. *European Journal of Communication*, *17*(1), 65–84. https://doi.org/10.1177/0267323102017001607

Cai, R., Cain, L. N., & Jeon, H. (2022). Customers' perceptions of hotel AI-enabled voice assistants: Does brand matter? *International Journal of Contemporary Hospitality Management*, *34*(8), 2807–2831. https://doi.org/10.1108/IJCHM-10-2021-1313

Cascella, M., Montomoli, J., Bellini, V., & Bignami, E. (2023). Evaluating the feasibility of ChatGPT in healthcare: An analysis of multiple clinical and research scenarios. *Journal of Medical Systems*, *47*(1), 33–33. https://doi.org/10.1007/s10916-023-01925-4

Chatterjee, S., Chaudhuri, R., Vrontis, D., Thrassou, A., & Ghosh, S. K. (2021). Adoption of artificial intelligence-integrated CRM systems in agile organizations in India. *Technological Forecasting and Social Change*, *168*, 120783. https://doi.org/10.1016/j.techfore.2021.120783

Choe, J. Y., Kim, J. J., & Hwang, J. (2022). Innovative robotic restUSErants in Korea: Merging a technology acceptance model and theory of planned behaviour. *Asian Journal of Technology Innovation*, *30*(2), 466–489. https://doi.org/10.1080/19761597.2021.2005466

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, *13*(3), 319–340. https://doi.org/10.2307/249008

Davis, F. D., Bagozzi, R. P., & Warshaw, P. R. (1989). User acceptance of computer technology: A comparison of two theoretical models. *Management Science*, *35*(8), 982–1003. https://doi.org/10.1287/mnsc.35.8.982

De Graaf, M. M., & Allouch, S. B. (2013). Exploring influencing variables for the acceptance of social robots. *Robotics and Autonomous Systems*, *61*(12), 1476–1486. https://doi.org/10.1016/j.robot.2013.07.007

Degen, H., & Ntoa, S. (2022). Exploring the design context of AI-powered services: A qualitative investigation of designers' experiences with machine learning. In *Artificial Intelligence in HCI* (Vol. 13336, pp.3–21). Springer International Publishing AG. https://doi.org/10.1007/978-3-031-05643-7_1

Etemad-Sajadi, R., & Sturman, M. C. (2022). How to increase the customer experience by the usage of remote control robot concierge solutions. *International Journal of Social Robotics*, *14*(2), 429–440. https://doi.org/10.1007/s12369-021-00800-x

Fernandes, T., & Oliveira, E. (2021). Understanding consumers' acceptance of automated technologies in service encounters: Drivers of digital voice assistants adoption. *Journal of Business Research*, *122*, 180–191. https://doi.org/10.1016/j.jbusres.2020.08.058

Forgas-Coll, S., Huertas-Garcia, R., Andriella, A., & Alenyà, G. (2022). How do consumers' gender and rational thinking affect the acceptance of entertainment social robots? *International Journal of Social Robotics*, *14*(4), 973–994. https://doi.org/10.1007/s12369-021-00845-y

Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, *18*(1), 39–50. https://doi.org/10.2307/3151312

Freeman, J., Park, S., & Middleton, C. (2020). Technological literacy and interrupted internet access. *Information, Communication & Society*, *23*(13), 1947–1964. https://doi.org/10.1080/1369118X.2019.1623901

Haggstrom, D. A., Lee, J. L., Dickinson, S. L., Kianersi, S., Roberts, J. L., Teal, E., Baker, L. B., & Rawl, S. M. (2019). Rural and urban differences in the adoption of new health information and medical technologies. *Journal of Rural Health: Official Journal of the American Rural Health Association and the National Rural Health Care Association*, *35*(2), 144–154. https://doi.org/10.1111/jrh.12358

Hair, Jr., J. F., Sarstedt, M., Matthews, L. M., & Ringle, C. M. (2016). Identifying and treating unobserved heterogeneity with FIMIX-PLS: part I–method. *European Business Review*, *28*(1), 63–76.

Haque, M. U., Dharmadasa, I., Sworna, Z. T., Rajapakse, R. N., & Ahmad, H. (2022). "I think this is the most disruptive technology":

Exploring Sentiments of ChatGPT Early Adopters using Twitter Data. arXiv preprint arXiv:2212.05856.
Hargittai, E., & Hinnant, A. (2008). Digital inequality: Differences in young adults' use of the internet. *Communication Research*, *35*(5), 602–621. https://doi.org/10.1177/0093650208321782
Hu, P. J., Chau, P. Y. K., Sheng, O. R. L., & Tam, K. Y. (1999). Examining the technology acceptance model using physician acceptance of telemedicine technology. *Journal of Management Information Systems*, *16*(2), 91–112. https://doi.org/10.1080/07421222.1999.11518247
Jamei, E., Mortimer, M., Seyedmahmoudian, M., Horan, B., & Stojcevski, A. (2017). Investigating the role of virtual reality in planning for sustainable smart cities. *Sustainability*, *9*(11), 2006. https://doi.org/10.3390/su9112006
Jo, H. (2023a). Understanding AI tool engagement: A study of ChatGPT usage and word-of-mouth among university students and office workers. *Telematics and Informatics*, *85*, 102067. https://doi.org/10.1016/j.tele.2023.102067
Jo, H. (2023b). Decoding the ChatGPT mystery: A comprehensive exploration of factors driving AI language model adoption. *Information Development*, 02666669231202764. https://doi.org/10.1177/02666669 231202764
Jo, H., & Park, D.-H. (2023). AI in the workplace: Examining the effects of ChatGPT on information support and knowledge acquisition. *International Journal of Human–Computer Interaction*, 1–16. https://doi.org/10.1080/10447318.2023.2278283
Klaus, P., & Zaichkowsky, J. L. (2022). The convenience of shopping via voice AI: Introducing AIDM. *Journal of Retailing and Consumer Services*, *65*, 102490. https://doi.org/10.1016/j.jretconser.2021.102490
Kline, R. B. (2011). *Principles and practice of structural equation modeling*. Guilford Press.
Koc, E., Hatipoglu, S., Kivrak, O., Celik, C., & Koc, K. (2023). Houston, we have a problem!: The use of ChatGPT in responding to customer complaints. *Technology in Society*, *74*, 102333. https://doi.org/10.1016/j.techsoc.2023.102333
Kuang, Y.-R., Zou, M.-X., Niu, H.-Q., Zheng, B.-Y., Zhang, T.-L., & Zheng, B.-W. (2023). ChatGPT encounters multiple opportunities and challenges in neurosurgery. *International Journal of Surgery (London, England)*, *109*(10), 2886–2891. https://doi.org/10.1097/JS9.0000000000000571
Lam, L. W. (2012). Impact of competitiveness on salespeople's commitment and performance. *Journal of Business Research*, *65*(9), 1328–1334. https://doi.org/10.1016/j.jbusres.2011.10.026
Legris, P., Ingham, J., & Collerette, P. (2003). Why do people use information technology? A critical review of the technology acceptance model. *Information & Management*, *40*(3), 191–204. https://doi.org/10.1016/S0378-7206(01)00143-4
Litt, E. (2013). Measuring users' internet skills: A review of past assessments and a look toward the future. *New Media & Society*, *15*(4), 612–630. https://doi.org/10.1177/1461444813475424
Liu, X., He, X., Wang, M., & Shen, H. (2022). What influences patients' continuance intention to use AI-powered service robots at hospitals? The role of individual characteristics. *Technology in Society*, *70*, 101996. https://doi.org/10.1016/j.techsoc.2022.101996
Liu, G., & Ma, C. (2023). Measuring EFL learners' use of ChatGPT in informal digital learning of English based on the technology acceptance model. *Innovation in Language Learning and Teaching*, 1–14. https://doi.org/10.1080/17501229.2023.2240316
Liu, K., & Tao, D. (2022). The roles of trust, personalization, loss of privacy, and anthropomorphism in public acceptance of smart healthcare services. *Computers in Human Behavior*, *127*, 107026. https://doi.org/10.1016/j.chb.2021.107026
Liu, H., Wang, L., & Koehler, M. J. (2019). Exploring the intention-behavior gap in the technology acceptance model: A mixed-methods study in the context of foreign-language teaching in China. *British Journal of Educational Technology*, *50*(5), 2536–2556. https://doi.org/10.1111/bjet.12824
Lu, L., Cai, R., & Gursoy, D. (2019). Developing and validating a service robot integration willingness scale. *International Journal of Hospitality Management*, *80*, 36–51. https://doi.org/10.1016/j.ijhm.2019.01.005
Lythreatis, S., Singh, S. K., & El-Kassar, A.-N. (2022). The digital divide: A review and future research agenda. *Technological Forecasting and Social Change*, *175*, 121359. https://doi.org/10.1016/j.techfore.2021.121359
M. Mijwil, M., Hiran, K. K., Doshi, R., Dadhich, M., Al-Mistarehi, A.-H., & Bala, I. (2023). ChatGPT and the future of academic integrity in the artificial intelligence era: A new frontier. *Al-Salam Journal for Engineering and Technology*, *2*(2), 116–127. https://doi.org/10.55145/ajest.2023.02.02.015
Maad, M., & Aljanabi, M, ChatGPT. (2023). Towards artificial intelligence-based cybersecurity: The practices and ChatGPT generated ways to combat cybercrime. *Iraqi Journal for Computer Science and Mathematics*, *4*(1), 65–70. https://doi.org/10.52866/ijcsm.2023.01.01.0019
Mailizar, M., Burg, D., & Maulina, S. (2021). Examining university students' behavioural intention to use e-learning during the COVID-19 pandemic: An extended TAM model. *Education and Information Technologies*, *26*(6), 7057–7077. https://doi.org/10.1007/s10639-021-10557-5
Malik, T., Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., Baabdullah, A. M., Koohang, A., Raghavan, V., Ahuja, M., Albanna, H., Albashrawi, M. A., Al-Busaidi, A. S., Balakrishnan, J., Barlette, Y., Basu, S., Bose, I., Brooks, L., Buhalis, D., … Wright, R. (2023). "So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management*, *71*, 102642. https://doi.org/10.1016/j.ijinfomgt.2023.102642
Marangunic, N., & Granic, A. (2015). Technology acceptance model: A literature review from 1986 to 2013. *Universal Access in the Information Society*, *14*(1), 81–95. https://doi.org/10.1007/s10209-014-0348-1
Marshall, A., Turner, K., Richards, C., Foth, M., & Dezuanni, M. (2022). Critical factors of digital AgTech adoption on Australian farms: From digital to data divide. *Information, Communication & Society*, *25*(6), 868–886. https://doi.org/10.1080/1369118X.2022.2056712
Menon, D., & Shilpa, K. (2023). "Chatting with ChatGPT": Analyzing the factors influencing users' intention to use the Open AI's ChatGPT using the UTAUT model. *Heliyon*, *9*(11), e20962–e20962. https://doi.org/10.1016/j.heliyon.2023.e20962
Mohr, N., & Kühl, R. S. (2021). Acceptance of artificial intelligence in German agriculture: An application of the technology acceptance model and the theory of planned behavior. *Precision Agriculture*, *22*(6), 1816–1844. https://doi.org/10.1007/s11119-021-09814-x
Montes, G. A., & Goertzel, B. (2019). Distributed, decentralized, and democratized artificial intelligence. *Technological Forecasting and Social Change*, *141*, 354–358. https://doi.org/10.1016/j.techfore.2018.11.010
Moon, J.-W., & Kim, Y.-G. (2001). Extending the TAM for a World-Wide-Web context. *Information & Management*, *38*(4), 217–230. https://doi.org/10.1016/S0378-7206(00)00061-6
Nguyen, T. D., Nguyen, D. T., & Cao, T. H. (2014). *Acceptance and use of information system: E-learning based on cloud computing in Vietnam*. In Information and Communication Technology: Second IFIP TC5/8 International Conference, ICT-EurAsia 2014, Bali, Indonesia, April 14–17, 2014; Proceedings 2 (pp. 139–149). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-55032-4_14
Ong, C.-S., & Lai, J.-Y. (2006). Gender differences in perceptions and relationships among dominants of e-learning acceptance. *Computers in Human Behavior*, *22*(5), 816–829. https://doi.org/10.1016/j.chb.2004.03.006
Oviedo-Trespalacios, O., Peden, A. E., Cole-Hunter, T., Costantini, A., Haghani, M., Rod, J. E., Kelly, S., Torkamaan, H., Tariq, A., David Albert Newton, J., Gallagher, T., Steinert, S., Filtness, A. J., & Reniers, G. (2023). The risks of using ChatGPT to obtain common safety-related information and advice. *Safety Science*, *167*, 106244. https://doi.org/10.1016/j.ssci.2023.106244
Pillai, R., & Sivathanu, B. (2020). Adoption of AI-based chatbots for hospitality and tourism. *International Journal of Contemporary Hospitality Management*, *32*(10), 3199–3226. https://doi.org/10.1108/IJCHM-04-2020-0259

Pillai, R., Sivathanu, B., Metri, B., & Kaushik, N. (2023). Students' adoption of AI-based teacher-bots (T-bots) for learning in higher education. *Information Technology & People*, *37*(1), 328–355. https://doi.org/10.1108/ITP-02-2021-0152

Porto, N., Huang, Y., & Xiao, J. J. (2019). Credit card adoption and usage in china: Urban–rural comparisons. *Singapore Economic Review*, *64*(01), 41–56. https://doi.org/10.1142/S021759081743010X

Quan, S. J. (2022). Urban-GAN: An artificial intelligence-aided computation system for plural urban design. *Environment and Planning B: Urban Analytics and City Science*, *49*(9), 2500–2515. https://doi.org/10.1177/23998083221100550

Quan, W., Moon, H., Kim, S. S., & Han, H. (2023). Mobile, traditional, and cryptocurrency payments influence consumer trust, attitude, and destination choice: Chinese versus Koreans. *International Journal of Hospitality Management*, *108*, 103363. https://doi.org/10.1016/j.ijhm.2022.103363

Rahimzadeh, V., Kostick-Quenet, K., Blumenthal Barby, J., & McGuire, A. L. (2023). Ethics education for healthcare professionals in the era of ChatGPT and other large language models: Do we still need it? *American Journal of Bioethics: AJOB*, *23*(10), 17–27. https://doi.org/10.1080/15265161.2023.2233358

Riddell, W. C., & Song, X. (2017). The role of education in technology use and adoption: Evidence from the Canadian workplace and employee survey. *ILR Review*, *70*(5), 1219–1253. https://doi.org/10.1177/0019793916687719

Scherer, R., Siddiq, F., & Tondeur, J. (2019). The technology acceptance model (TAM): A meta-analytic structural equation modeling approach to explaining teachers' adoption of digital technology in education. *Computers & Education*, *128*, 13–35. https://doi.org/10.1016/j.compedu.2018.09.009

Serenko, A. (2008). A model of user adoption of interface agents for email notification. *Interacting with Computers*, *20*(4–5), 461–472. https://doi.org/10.1016/j.intcom.2008.04.004

Shah, V., & Kwak, N. (2001). "Connecting" and "disconnecting" with civic life: Patterns of internet use and the production of social capital. *Political Communication*, *18*(2), 141–162. https://doi.org/10.1080/105846001750322952

Shaikh, A. A., & Karjaluoto, H. (2015). Mobile banking adoption: A literature review. *Telematics and Informatics*, *32*(1), 129–142. https://doi.org/10.1016/j.tele.2014.05.003

Sharma, S., Singh, G., Gaur, L., & Afaq, A. (2022). Exploring customer adoption of autonomous shopping systems. *Telematics and Informatics*, *73*, 101861. https://doi.org/10.1016/j.tele.2022.101861

Shoufan, A. (2023). Exploring students' perceptions of ChatGPT: Thematic analysis and follow-up survey. *IEEE Access*, *11*, 38805–38818. https://doi.org/10.1109/ACCESS

Stahl, B. C., & Eke, D. (2024). The ethics of ChatGPT – Exploring the ethical issues of an emerging technology. *International Journal of Information Management*, *74*, 102700. https://doi.org/10.1016/j.ijinfomgt.2023.102700

Tandon, A., Sharma, H., & Aggarwal, A. G. (2019). Studying the moderating effect of a respondent's locality in M-commerce adoption intention. *Ingeniería Solidaria*, *15*(29), 1–23. https://doi.org/10.16925/2357-6014.2019.03.07

Taylor, A., & Dalal, H. A. (2017). Gender and information literacy: Evaluation of gender differences in a student survey of information sources. *College & Research Libraries*, *78*(1), 90–113. https://doi.org/10.5860/crl.78.1.90

Terzis, V., & Economides, A. A. (2011). Computer based assessment: Gender differences in perceptions and acceptance. *Computers in Human Behavior*, *27*(6), 2108–2122. https://doi.org/10.1016/j.chb.2011.06.005

Uludag, K. (2023). The use of AI-supported Chatbot in Psychology. Available at SSRN 4331367.

van Dis, E. A. M., Bollen, J., Zuidema, W., van Rooij, R., & Bockting, C. L. (2023). ChatGPT: Five priorities for research. *Nature (London)*, *614*(7947), 224–226. https://doi.org/10.1038/d41586-023-00288-7

Venkatesh, V., & Bala, H. (2008). Technology acceptance model 3 and a research agenda on interventions. *Decision Sciences*, *39*(2), 273–315. https://doi.org/10.1111/j.1540-5915.2008.00192.x

Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. *Management Science*, *46*(2), 186–204. https://doi.org/10.1287/mnsc.46.2.186.11926

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, *27*(3), 425–478. https://doi.org/10.2307/30036540

Vimalkumar, M., Sharma, S. K., Singh, J. B., & Dwivedi, Y. K. (2021). "Okay Google, what about my privacy?": User's privacy perceptions and acceptance of voice based digital assistants. *Computers in Human Behavior*, *120*, 106763. https://doi.org/10.1016/j.chb.2021.106763

Wang, C., Ahmad, S. F., Bani Ahmad Ayassrah, A. Y. A., Awwad, E. M., Irshad, M., Ali, Y. A., Al-Razgan, M., Khan, Y., & Han, H. (2023). An empirical evaluation of technology acceptance model for artificial intelligence in E-commerce. *Heliyon*, *9*(8), e18349–e18349. https://doi.org/10.1016/j.heliyon.2023.e18349

Wu, M., Wang, N., & Yuen, K. F. (2023). Can autonomy level and anthropomorphic characteristics affect public acceptance and trust towards shared autonomous vehicles? *Technological Forecasting and Social Change*, *189*, 122384. https://doi.org/10.1016/j.techfore.2023.122384

Yang, D. (2023). Income growth, income uncertainty, and urban–rural household savings rate in China. *Regional Science and Urban Economics*, *99*, 103855. https://doi.org/10.1016/j.regsciurbeco.2022.103855

Yang, F., & Zhang, X. (2020). Focal fields in literature on the information divide: The USA, China, UK and India. *Journal of Documentation*, *76*(2), 373–388. https://doi.org/10.1108/JD-02-2019-0032

Yao Tian, T., Lik-Hang, C., Reza Hadi Mogavi, L., Liao, Y., & Zhou, P. (2023). Last week with ChatGPT: A Weibo study on social perspective regarding ChatGPT for education and beyond. ArXiv.Org. https://doi.org/10.48550/arxiv.2306.04325

Yu, X., & Liu, S. (2022). Disparities in online use behaviours and Chinese digital inclusion: A 10-year comparison. *International Journal of Environmental Research and Public Health*, *19*(19), 11937. https://doi.org/10.3390/ijerph191911937

Zhang, B., Zhu, Y., Deng, J., Zheng, W., Liu, Y., Wang, C., & Zeng, R. (2023). "I am here to assist your tourism": Predicting continuance intention to use AI-based chatbots for tourism. Does gender really matter? *International Journal of Human–Computer Interaction*, *39*(9), 1887–1903. https://doi.org/10.1080/10447318.2022.2124345

About the authors

Jiaojiao Ma, PhD Candidate at the School of Media and Communication, Shanghai Jiao Tong University. Her research interests include smart media use and new media technologies.

Pengcheng Wang, Assistant Professor at the School of Media and Communication, Shanghai Jiao Tong University. His main research interests include cyberpsychology.

Benqian Li, Dean and Professor at the School of Media and Communication, Shanghai Jiao Tong University.

Tian Wang, Master Degree Candidate at the School of Media and Communication, Shanghai Jiao Tong University. Her research interests include communication psychology, intelligent communication, and consumer behavior.

Xiang Shan Pang, Master Degree Candidate at the School of Media and Communication, Shanghai Jiao Tong University. Her research interests include new media and bilingual communication studies.

Dake Wang, Associate researcher at the School of Media and Communication, Shanghai Jiao Tong University. His recent research project centers on science communication and social impact of new media development.

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